

# Jeeho Shin

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## EDUCATION

### Columbia University

MS in Computer Science (Machine Learning Track)

New York, NY

Sep 2026 – Dec 2027

### Korea Advanced Institute of Science and Technology (KAIST)

BS in Computer Science

Daejeon, KR

Aug 2026

Courses: Machine Learning, Computer Vision, Information Retrieval, Database, Data Structures, Algorithms

## SKILLS

**Programming Languages:** Python, Java, SQL, C, TypeScript

**Machine Learning & AI:** PyTorch, scikit-learn, Transformers, TRL, LangChain, LangGraph, Langfuse, LangSmith, W&B

**Frameworks & Tools:** FastAPI, React, Supabase, PostgreSQL, Docker, Vercel, Fly.io, pytest, Vitest, GitHub Actions, Jira

**Languages:** Korean, English

## EXPERIENCE

### AI Research Intern

Seongnam, KR

NC AI (AI Research Organization of NCSOFT) – Chatbot Service Team

Sep 2025 – Feb 2026

- Designed evaluation system for LangGraph-based RAG chatbot to gauge response quality without relying on external LLM API calls, mitigating data privacy risks by fine-tuning open-source LLM with GRPO method
- Implemented context-aware decoding pipeline with Transformers library to generate synthetic responses with varying contextual relevance, strengthening model's response-ranking capability via custom reward functions
- Validated fine-tuned model on 500 internal chat samples, achieving 86.3% agreement with human evaluators, 8.8% improvement over standard RAGAS metrics, while cutting evaluation latency by 20%
- Deployed fine-tuned model as Docker container and integrated evaluation module as LangGraph node, enabling real-time, privacy-preserving response quality checks through Langfuse within live chatbot workflow
- Built automated alerting flow that posts low-scoring responses to Chatbot Service Team's Microsoft Teams channel as Adaptive Cards via Teams webhook, providing engineers with visibility into production failures

### Undergraduate Researcher

Seoul, KR

KAIST AI – Data Mining Lab

Jan 2025 – Aug 2025

- Identified underutilization of user reviews as preference signal in multimodal recipe recommenders and developed framework incorporating users' dietary preferences into user and recipe representation learning
- Constructed data pipeline across 2 datasets (~76K users, ~75K recipes), prompting VLM and LLM to summarize each user's dietary preference and each recipe's image, ingredients, and nutrition information
- Encoded these summaries into multimodal embeddings and applied graph propagation and cross-view contrastive learning to align multimodal and learnable user and recipe embeddings
- Achieved 7–15% Recall@10 gains over 3 advanced multimodal recipe recommendation baselines, reducing training time and peak VRAM usage for learning user and recipe embeddings to one-third

## PROJECTS

### LionPlanner ([Link](#))

- Built full-stack course-planning web app end-to-end, combining React/TypeScript frontend, LangGraph-based agentic chatbot, FastAPI backend, and Supabase vector search, deployed to production on Vercel and Fly.io
- Scraped syllabi and reviews for 36 Columbia CS courses and created workload-grading pipeline across 5 dimensions using LLM, grounding each score in verbatim evidence with anchor calibration
- Engineered agentic workflow generating responses by calling tools for semantic search, degree-requirement lookup, and schedule-conflict checking to deliver context-aware course recommendations
- Evaluated recommendation quality on 100 queries with LLM evaluator (avg. 4.35/5), tracing each tool's output step by step through LangSmith to pinpoint where failures most often occurred
- Set up GitHub Actions CI pipeline that runs backend and frontend tests with pytest and Vitest on every push

## PUBLICATIONS

- From Raw Features to Effective Embeddings: A Three-Stage Approach for Multimodal Recipe Recommendation  
*Jeeho Shin, Kyungho Kim, and Kijung Shin, SIGIR 2026, [Link](#)*
- Large language models are zero-shot point-of-interest recommenders  
*Joeun Kim, Youngjin Seo, Yeonsoo Kim, Junhyeok Kang, Jeeho Shin, Patara Trirat, and Jae-Gil Lee, DMKD 2025, [Link](#)*